

# G. ARAVIND

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## EDUCATION

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|------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Indian Institute of Science, Bangalore</b><br>M.Tech in Artificial Intelligence             | Oct 2020 – Present<br>8.1/10    |
| <b>Heritage Institute of Technology, Kolkata</b><br>B.Tech in Computer Science and Engineering | Aug 2016 – Jul 2020<br>8.8 / 10 |

## RELEVANT COURSEWORK

**Foundational Courses:** Pattern Recognition and Neural Networks, Data Structures, Computer Vision, Data Analytics  
**Advanced Courses:** Reinforcement Learning, Advanced Deep Learning, Deep Learning for Natural Language Processing  
**Math Courses:** Random Processes, Computational Linear Algebra, Computational Methods of Optimization  
**Online Courses:** Machine Learning with Graphs (Stanford)

## WORK EXPERIENCE

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| <b>Mastercard</b>   Data Scientist Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jun 2021 – Jul 2021 |
| <ul style="list-style-type: none"><li>– Interned with AI Garage team at Mastercard, which focuses on designing AI-based solutions for use-cases in payments</li><li>– Majority of my time as an intern was devoted to building models for predicting high-value financial transactions, and the time and category of future transactions for a customer</li><li>– Modeled transactions as Marked Temporal Point Processes and used it to make predictions about future payments</li></ul> |                     |

## ACADEMIC PROJECTS

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| <b>Adversarial Attacks on Recommendation System Models</b>   <i>Thesis Project</i>   Prof. Shirish Shevade                                                                                                                                                                                                                                                                                                                                                                                                      | (ongoing)  |
| <ul style="list-style-type: none"><li>– Using hyperparameter tuning techniques to tune the graph data in order to drop the recommender's performance</li><li>– Attacking the user-item graph using meta-learning techniques, to reduce performance of collaborative filtering systems</li></ul>                                                                                                                                                                                                                 |            |
| <b>Multimodal Transformers for Image Captioning</b> 🔗   Prof. Sriram Ganapathy                                                                                                                                                                                                                                                                                                                                                                                                                                  | (Sep 2021) |
| <ul style="list-style-type: none"><li>– Image captioning formulated as a multimodal translation task, where input image features are extracted using various encoder models, and fed to deep networks for caption generation</li><li>– Used multimodal Transformer to capture both intra-modal and inter-modal interactions in a unified attention block</li></ul>                                                                                                                                              |            |
| <b>C-LSTMs vs Transformers for Multi-class Text Classification</b> 🔗   Prof. Shirish Shevade                                                                                                                                                                                                                                                                                                                                                                                                                    | (Aug 2021) |
| <ul style="list-style-type: none"><li>– Implemented C-LSTM architecture (a combination of CNN and deep LSTMs) for classifying text documents</li><li>– Used self-attention mechanism at output, and dynamic meta-embeddings at input to improve C-LSTM performance</li><li>– Also used Transformers for document classification, and experimented with encoder blocks and positional embeddings</li><li>– Used Deep Averaging Networks (DANs) as baseline and tried different word/bi-gram embeddings</li></ul> |            |

## PERSONAL PROJECTS

- Modeling the spread of COVID-19 in Karnataka, India 🔗
- Movie recommendation system using graph neural networks and collaborative filtering 🔗
- Dynamic meta-embeddings for improved sentence representations 🔗
- Duckworth-Lewis method for predicting target runs in cricket matches, using non-linear regression models 🔗
- Gradient based optimization of hyperparameters for linear models 🔗
- Image compression using principal component analysis 🔗

## TECHNICAL SKILLS

**Languages:** Python, C, MATLAB, SQL

**Technologies / Frameworks:** PyTorch, TensorFlow, Scikit-Learn

## ACHIEVEMENTS / EXTRACURRICULAR

- All India Rank 6<sup>th</sup> in GATE Computer Science exam, out of ~1,00,000 candidates
- Teaching Assistant for E2 202: Random Processes, Fall 2021
- Certified in "CODECHEF Certified Data Structures and Algorithms Programme", Advanced Level (Jan 2018)